PKG 09: COMPARISON 01 and JAVA LEXICOGRAPHY

- 8 Primitives: boolean, byte, char, short, int, long, float, and double
- 6 Relational Operators / Comparison Operators / Ordering Operators

```
1  System.out.println(1 == 1); // true
2  System.out.println(1 != 1); // false
3  System.out.println(2 > 2);  // false
4  System.out.println(2 >= 2); // true
5  System.out.println(3 < 3);  // false
6  System.out.println(3 <= 3); // true
7
8  System.out.println(true == true);   // true
9  System.out.println(false != false); // false
```

```
10 System.out.println(10 == 10);
11 System.out.println(10 == 15);
12 System.out.println('a' == 97.0);
13 System.out.println((int)'a');       // ASCII Value
```

```
14 String courseA = new String("CSC220");
15 String courseB = new String("CSC220");
16 if (courseA == courseB) {
17     System.out.println("Yes, equal");
18 } else {
19     System.out.println("No, unequal");
20 }
21 if (courseA.equals(courseB)) {
22     System.out.println("Yes, equal");
23 } else {
24     System.out.println("No, unequal");
25 }
```

```
26 String name = "Mickey Mouse character".substring(0, 12);
27 if (name == "Mickey Mouse") {
28     System.out.println("Yes, equal");
29 } else {
30     System.out.println("No, unequal");
31 }
32 if (name.equals("Mickey Mouse")) {
33     System.out.println("Yes, equal");
34 } else {
35     System.out.println("No, unequal");
36 }
```

Not equal because they are comparing an object and a literal string, they have nothing to do with each other.

```
37 String[] strArr1 = {"Java", "C", "C++", "Python"};
38 String[] strArr2 = {"Java", "C", "C++", "Python"};
39
40 System.out.println("Compare contents");
41 System.out.println(Arrays.equals(strArr1, strArr2));
42
43 System.out.println("Compare references");
44 System.out.println(strArr1 == strArr1);
45 System.out.println(strArr1.equals(strArr1));
46
47 System.out.println("Compare references");
48 System.out.println(strArr1 == strArr2);
49 System.out.println(strArr1.equals(strArr2));
```

ONLY works for 1D array
method deepEquals() for 2D array

58. they are comparing references not the content

61, 62 FALSE because they are two different objects

COMPARISON 01 and JAVA LEXICOGRAPHY

```
        String a = "a", b = "b";

        // Java lexicography
        a = "abc";                          // same chars at
        b = "abc";                          // same positions

        System.out.println(b.compareTo(a)); // 0

        a = "abc";                          // length = 3
        b = "abcdefg";                      // length = 7

        System.out.println(b.compareTo(a)); // 4

        a = "abcd";                         // ASCII d = 100
        b = "abcz";                         // ASCII z = 122

        System.out.println(b.compareTo(a)); // 22

        a = "aAadzzza";                     // ASCII A = 65
        b = "aaazdddddb";                   // ASCII a = 97

        System.out.println(b.compareTo(a));    // 32

        char c = 'C';
        // System.out.println(b.compareTo(c)); // ?

        // Error: incompatible types: char cannot be converted to String
```

```
        String a, b;
        char c;

        // ASCII: F = 70, f = 102
        a = "Googlefi";
        b = "GoogleFI";

        System.out.println(a.compareTo(b));     // 32

        // ASCII:
        a = "GoogleDuplex";
        b = "GoogleDuplexAI";

        System.out.println(a.compareTo(b));     // -2

        // ASCII: X = 88, x = 120
        a = "GooglePixel3XL";
        b = "GooglePixel3xl";

        System.out.println(b.compareTo(a));     // 32

        // ASCII:
        a = "GooglePixel3XL";
        b = "GooglePixel3xl";

        System.out.println(a.compareTo(b));     // -32

        // ASCII: F = 70, G = 71, I = 73
        a = "F" + "I";
        c = 'G';
        // System.out.println(a.compareTo(c));  // Error
```